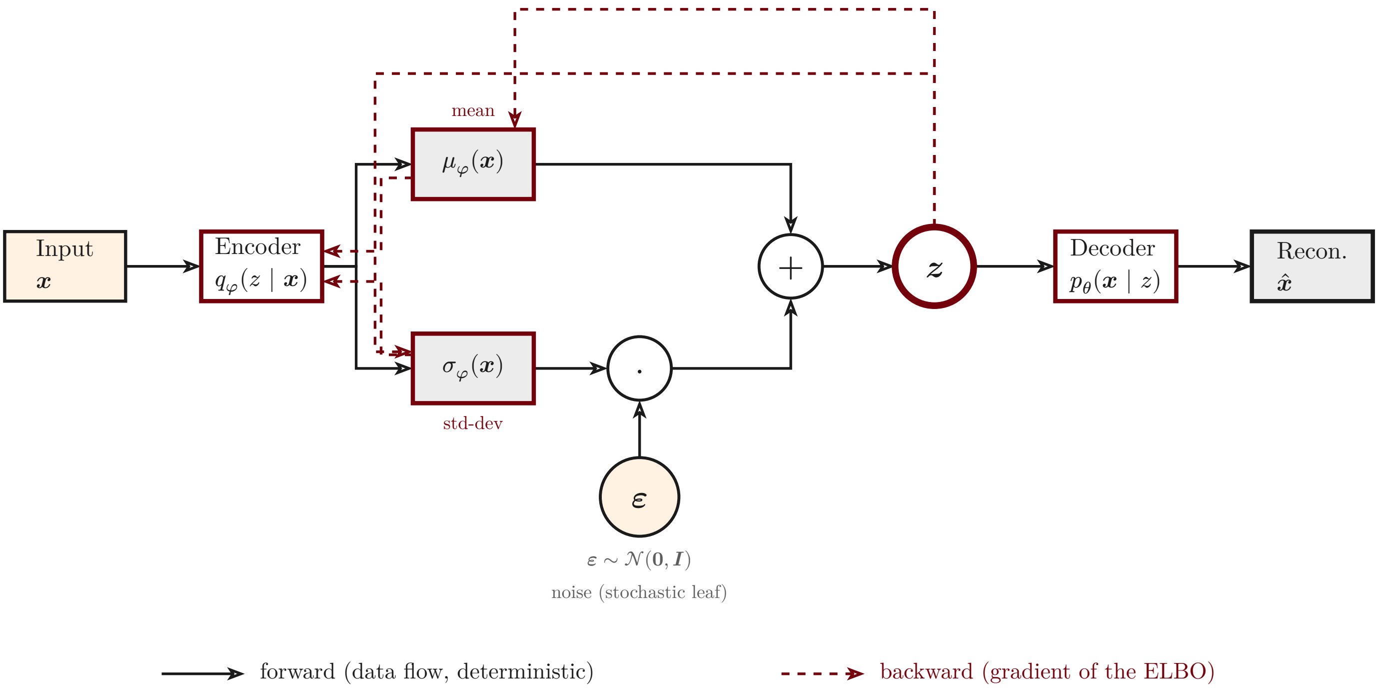
Reparameterization trick

$$z = \mu_\varphi(x) + \sigma_\varphi(x) \cdot \varepsilon, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$$



Sampling $z \sim \mathcal{N}(\mu, \sigma^2)$ directly is a **stochastic** node: no gradient can pass through a random draw. The trick moves the randomness into an auxiliary leaf $\varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and writes z as a **deterministic**, differentiable function of $(\mu, \sigma, \varepsilon)$. The backward pass (dashed) then flows through μ and σ into the encoder parameters φ — and simply **bypasses** the noise leaf ε.